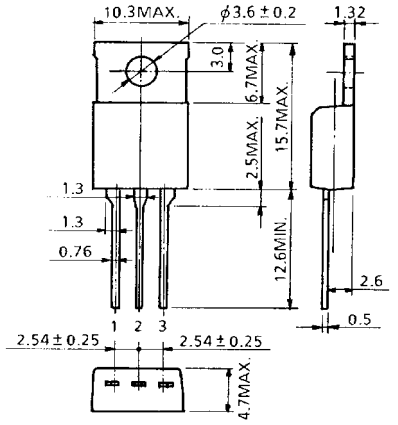
TOSHIBA BI−DIRECTIONAL TRIODE THYRISTOR SILICON PLANAR TYPE

SM6G45,SM6J45,SM6G45A,SM6J45A

AC POWER CONTROL APPLICATIONS

- Repetitive Peak Off−State Voltage : V_{DRM} = 400, 600V
- R.M.S ON−State Current : $I_{T\ (RMS)}$ = 6A
- High Commutating (dv / dt)

Unit: mm



JEDEC	TO−220AB
JEITA	—
TOSHIBA	13−10G1A

Weight: 2.0g

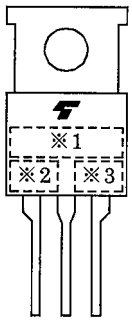
MAXIMUM RATINGS

CHARACTERISTIC		SYMBOL	RATING	UNIT
Repetitive Peak Off−State Voltage	SM6G45 SM6G45A	V_{DRM}	400	V
	SM6J45 SM6J45A		600	
R.M.S On−State Current (Full Sine Waveform Tc = 104°C)		$I_{T\ (RMS)}$	6	A
Peak One Cycle Surge On−State Current (Non−Repetitive)		I_{TSM}	60 (50Hz)	A
			66 (60Hz)	
I^2t Limit Value		I^2t	18	A^2s
Critical Rate of Rise of On−State Current		di / dt	50	A / μs
Peak Gate Power Dissipation		P_{GM}	5	W
Average Gate Power Dissipation		$P_{G\ (AV)}$	0.5	W
Peak Gate Voltage		V_{GM}	10	V
Peak Gate Current		I_{GM}	2	A
Junction Temperature		T_j	−40~125	°C
Storage Temperature Range		T_{stg}	−40~125	°C

ELECTRICAL CHARACTERISTICS (Ta = 25°C)

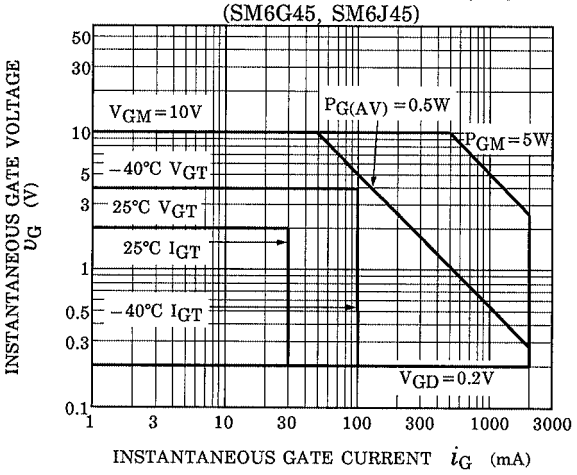
CHARACTERISTIC			SYMBOL	TEST CONDITION		MIN	TYP.	MAX	UNIT
Repetitive Peak Off−State Current			I_{DRM}	V_{DRM} = Rated, T_j = 125°C		—	—	2	mA
Gate Trigger Voltage	SM6G45 SM6J45	I	V_{GT}	V_D = 12V R_L = 20Ω	T2 (+), Gate (+)	—	—	2	V
		II			T2 (+), Gate (−)	—	—	2	
		III			T2 (−), Gate (−)	—	—	2	
		IV			T2 (−), Gate (+)	—	—	—	
	SM6G45A SM6J45A	I			T2 (+), Gate (+)	—	—	1.5	
		II			T2 (+), Gate (−)	—	—	1.5	
		III			T2 (−), Gate (−)	—	—	1.5	
		IV			T2 (−), Gate (+)	—	—	—	
Gate Trigger Current	SM6G45 SM6J45	I	I_{GT}	V_D = 12V R_L = 20Ω	T2 (+), Gate (+)	—	—	30	mA
		II			T2 (+), Gate (−)	—	—	30	
		III			T2 (−), Gate (−)	—	—	30	
		IV			T2 (−), Gate (+)	—	—	—	
	SM6G45A SM6J45A	I			T2 (+), Gate (+)	—	—	20	
		II			T2 (+), Gate (−)	—	—	20	
		III			T2 (−), Gate (−)	—	—	20	
		IV			T2 (−), Gate (+)	—	—	—	
Peak On−State Voltage			V_{TM}	I_{TM} = 9A		—	—	1.5	V
Gate Non−Trigger Voltage			V_{GD}	V_D = Rated, Tc = 125°C		0.2	—	—	V
Holding Current			I_H	V_D = 12V, I_{TM} = 1A		—	—	50	mA
Thermal Resistance			$R_{th\ (j-c)}$	Junction to Case, AC		—	—	2.5	°C / W
Critical Rate of Rise of Off−State Voltage at Commutation	SM6G45 SM6J45		(dv / dt) c	V_{DRM} = 400V, (di / dt) c = −3.3A / ms T_j = 125°C		10	—	—	V / μs
	SM6G45A SM6J45A					4	—	—	

MARKING

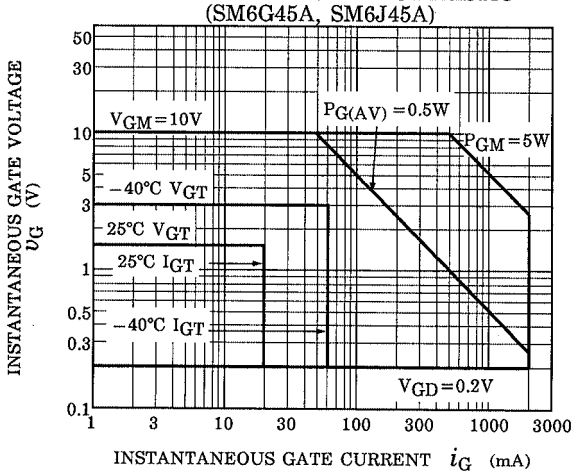


NUMBER	SYMBOL		MARK
* 1	TYPE	SM6G45, SM6G45A	M6G45
		SM6J45, SM6J45A	M6J45
* 2		SM6G45A, SM6J45A	A
* 3	Lot Number Month (Starting from Alphabet A) Year (Last Decimal Digit of the Current Year)		Example 8A : January 1998 8B : Febrary 1998 8L : December 1998

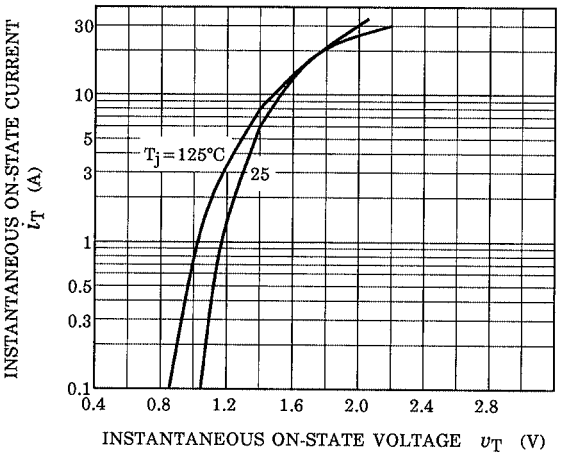
GATE TRIGGER CHARACTERISTIC
(SM6G45, SM6J45)



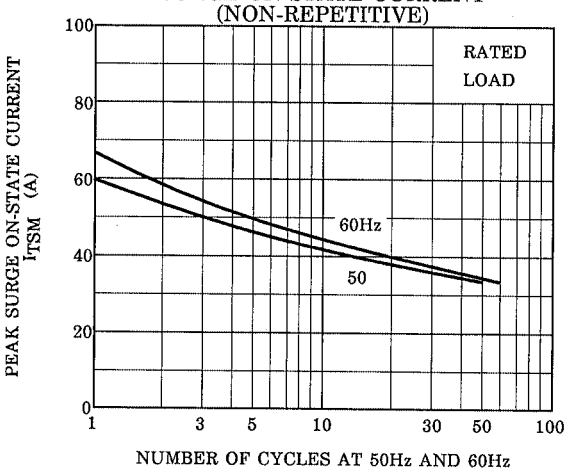
GATE TRIGGER CHARACTERISTIC
(SM6G45A, SM6J45A)



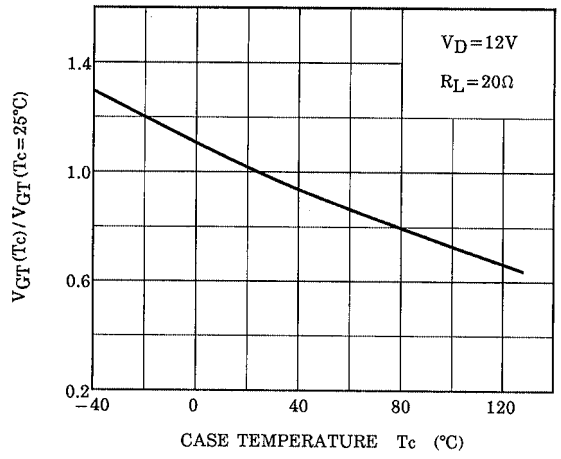
$i_T - v_T$



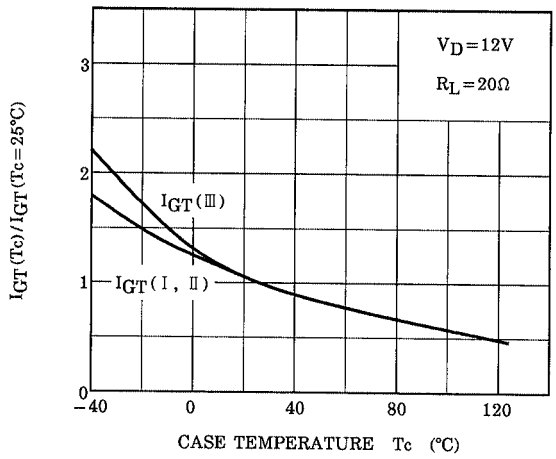
SURGE ON-STATE CURRENT
(NON-REPETITIVE)



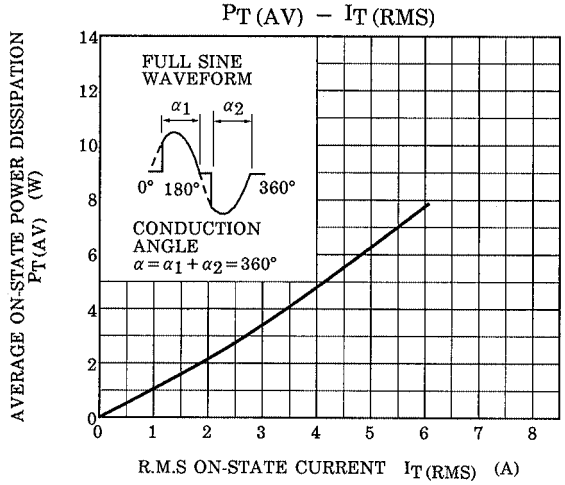
V_{GT} (Tc) / V_{GT} (Tc=25°C) − Tc (TYPICAL)



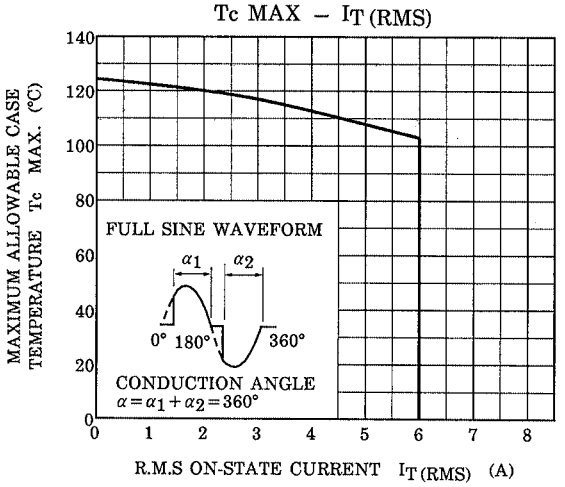
I_{GT} (Tc) / I_{GT} (Tc=25°C) − Tc (TYPICAL)



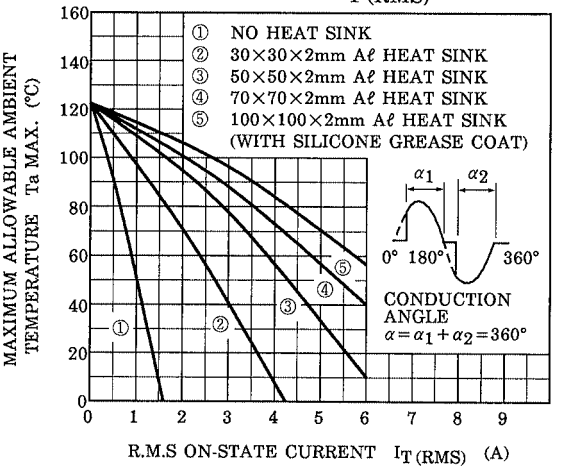
$P_{T(AV)}$ – $I_{T(RMS)}$



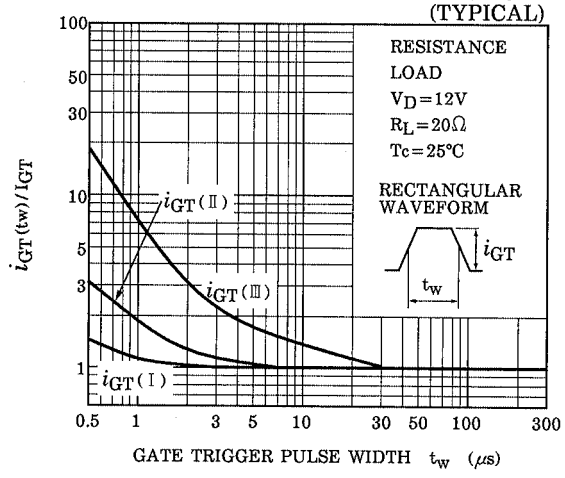
Tc MAX – $I_{T(RMS)}$



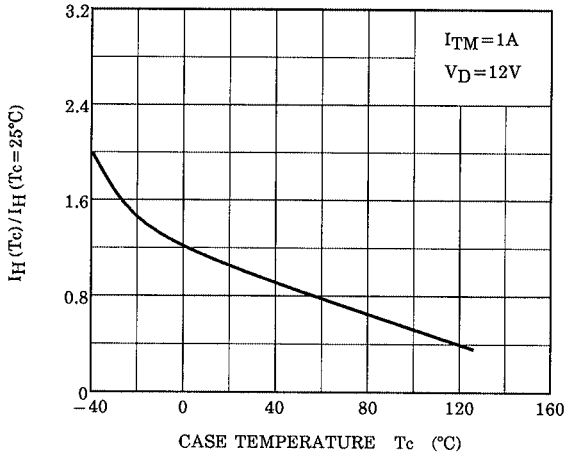
Ta MAX – $I_{T(RMS)}$



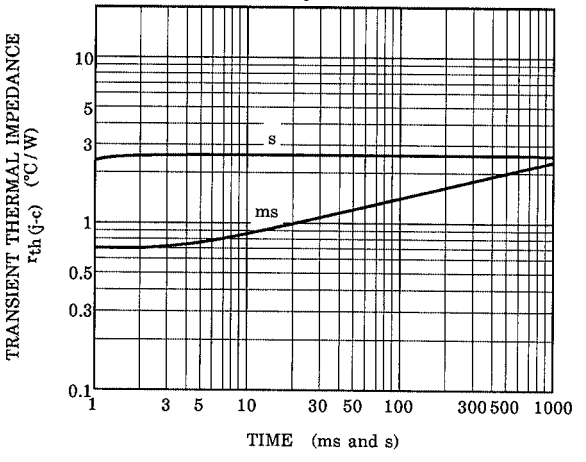
PULSE TRIGGER CHARACTERISTIC (TYPICAL)



I_H (Tc) / I_H (Tc=25°C) – Tc (TYPICAL)



$r_{th(j-c)}$ – t



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